Xiaoling Hu

E-mail: xihu3@mgh.harvard.edu, *Mobile*: 6312028413
Website: `https://huxiaoling.github.io/`

Current Position

- **Harvard Medical School, Athinoula A. Martinos Center for Biomedical Imaging, USA** Aug. 2023 - Present
  Postdoctoral Research Fellow
  - Hosted by Prof. Juan Eugenio Iglesias and Prof. Bruce Fischl

Research Interests

My research interests are in machine learning, computer vision, and their applications in healthcare, multimodal AI, generative AI, and trustworthy learning systems. In particular, I am interested in:

- **Topological Deep Learning**
- **Trustworthy Machine Learning**
- **Multimodal AI and Generative AI (GenAI)**
- **VLM/LLM Reasoning and Efficiency**

Education

- **Stony Brook University, Department of CS, USA** Jan. 2018 - June 2023
  Doctor of Philosophy
  - Advisor: Chao Chen
  – Thesis: Learning Topological Representations for Deep Image Understanding
  – Committee: Chao Chen, Dimitris Samaras, Haibin Ling, Li Fuxin

- **Tsinghua University, Department of EE, China** Sep. 2014 - June 2017
  Master of Engineering

- **Huazhong University of Science and Technology, Department of EE, China**
  Sep. 2010 - June 2014
  Bachelor of Science

Selected Publications

(* indicates equal contribution, __ denotes students (co-)mentored by me, ‡ denotes (co)-senior supervision)

[1] **Act Like a Pathologist: Tissue-Aware Whole Slide Image Reasoning**
Wentao Huang, Weimin Lyu, Peiliang Lou, Qingqiao Hu, **Xiaoling Hu**, Shahira Abousamra, Wenchao Han, Ruifeng Guo, Jiawei Zhou, Chao Chen, Chen Wang
*The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2026

[2] ***RankByGene*: Gene-Guided Histopathology Representation Learning Through Cross-Modal Ranking Consistency**
Wentao Huang, Meilong Xu, **Xiaoling Hu**, Shahira Abousamra, Aniruddha Ganguly, Saarthak Kapse, Alisa Yurovsky, Prateek Prasanna, Tahsin Kurc, Joel Saltz, Michael L. Miller, Chao Chen
Major revision for *IEEE Transactions on Medical Imaging (**TMI**)*

[3] **MATCH: Multi-faceted Adaptive Topo-Consistency for Semi-Supervised Histopathology Segmentation**
Meilong Xu*, **Xiaoling Hu***, Shahira Abousamra, Chen Li, Chao Chen
*Thirty-Ninth Conference on Neural Information Processing Systems (**NeurIPS**)*, 2025

Short version is selected as ***Oral Presentation*** *at Imageomics Workshop at NeurIPS 2025*

[4] **Text-Driven Weakly Supervised OCT Lesion Segmentation with Structural Guidance**
Jiaqi Yang, Nitish Mehta, **Xiaoling Hu**, Chao Chen, Chia-Ling Tsai
*The Journal of Biomedical and Health Informatics (**JBHI**)*, 2025

[5] ***Learn2Synth*: Learning Optimal Data Synthesis Using Hypergradients for Brain Image Segmentation**
Xiaoling Hu, Xiangrui Zeng, Oula Puonti, Juan Eugenio Iglesias, Bruce Fischl[‡], Yaël Balbastre[‡]
*International Conference on Computer Vision (**ICCV**)*, 2025

[6] ***TopoCellGen*: Generating Histopathology Cell Topology with a Diffusion Model**
Meilong Xu, Saumya Gupta, **Xiaoling Hu**, Chen Li, Shahira Abousamra, Dimitris Samaras, Prateek Prasanna, Chao Chen
*The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2025 (**Oral Presentation, acceptance rate < 1%**)
Short version is also presented at CVPR 2025 Workshop GMCV

[7] **Hierarchical Uncertainty Estimation for Learning-based Registration in Neuroimaging**
Xiaoling Hu, Karthik Gopinath, Peirong Liu, Malte Hoffmann, Koen Van Leemput, Oula Puonti[‡], Juan Eugenio Iglesias[‡]
*International Conference on Learning Representations (**ICLR**)*, 2025

[8] **Semi-supervised Segmentation of Histopathology Images with Noise-Aware Topological Consistency**
Meilong Xu, **Xiaoling Hu**, Saumya Gupta, Shahira Abousamra, Chao Chen
*European Conference on Computer Vision (**ECCV**)*, 2024

[9] **Brain-ID: Learning Contrast-agnostic Anatomical Representations for Brain Imaging**
Peirong Liu, Oula Puonti, **Xiaoling Hu**, Daniel C. Alexander, Juan Eugenio Iglesias
*European Conference on Computer Vision (**ECCV**)*, 2024

[10] **Registration by Regression (RbR): a framework for interpretable and flexible atlas registration**
Karthik Gopinath*, **Xiaoling Hu***, Malte Hoffmann, Oula Puonti[‡], Juan Eugenio Iglesias[‡]
*Workshop on Biomedical Image Registration-MICCAI (**WBIR**)*, 2024

[11] ***P-Count*: Persistence-based Counting of White Matter Hyperintensities in Brain MRI**
Xiaoling Hu, Annabel Sorby-Adams, Frederik Barkhof, William Kimberly, Oula Puonti, Juan Eugenio Iglesias
*Workshop on Topology- and Graph-Informed Imaging Informatics-MICCAI (**TGI3**)*, 2024

[12] **Semi-Supervised Contrastive VAE for Disentanglement of Digital Pathology Images**
Mahmudul Hasan, **Xiaoling Hu**, Shahira Abousamra, Prateek Prasanna, Joel Saltz, Chao Chen
*International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**)*, 2024

[13] **Hard Negative Sample Mining for Whole Slide Image Classification**
Wentao Huang, **Xiaoling Hu**, Shahira Abousamra, Prateek Prasanna, Chao Chen
*International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**)*, 2024

[14] **Spatial Diffusion for Cell Layout Generation**
Chen Li, **Xiaoling Hu**, Shahira Abousamra, Meilong Xu, Chao Chen
*International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**)*, 2024

[15] **Anomaly-guided weakly supervised lesion segmentation on retinal OCT images**
Jiaqi Yang, Nitish Mehta, Gozde Merve Demirci, **Xiaoling Hu**, Meera Ramakrishnan, Mina Naguib, Chao Chen, Chialing Tsai
*Medical Image Analysis (**MedIA**)*, 2024

[16] **Topology-Aware Uncertainty for Image Segmentation**
Saumya Gupta, Yikai Zhang, **Xiaoling Hu**, Prateek Prasanna, Chao Chen
*Thirty-Seventh Conference on Neural Information Processing Systems (**NeurIPS**)*, 2023

[17] **Calibrating Uncertainty for Semi-Supervised Crowd Counting**
Chen Li, **Xiaoling Hu**, Shahira Abousamra, Chao Chen
*International Conference on Computer Vision (**ICCV**)*, 2023

[18] **Enhancing Modality-Agnostic Representations via Meta-Learning for Brain Tumor Segmentation**
Aishik Konwer, **Xiaoling Hu**, Xuan Xu, Joseph Bae, Chao Chen, Prateek Prasanna
*International Conference on Computer Vision (**ICCV**)*, 2023

[19] **Learning Probabilistic Topological Representations Using Discrete Morse Theory**
Xiaoling Hu, Dimitris Samaras, Chao Chen
*International Conference on Learning Representations (**ICLR**)*, 2023 (**Spotlight Presentation, notable-top-25% of accepted papers**)
*Short version is selected as **Oral Presentation** at Medical Imaging meets NeurIPS Workshop, 2023*

[20] **Confidence Estimation Using Unlabeled Data**
Chen Li, **Xiaoling Hu**, Chao Chen
*International Conference on Learning Representations (**ICLR**)*, 2023

[21] **Structure-Aware Image Segmentation with Homotopy Warping**
Xiaoling Hu
*Thirty-Sixth Conference on Neural Information Processing Systems (**NeurIPS**)*, 2022

[22] **Learning Topological Interactions for Multi-Class Medical Image Segmentation**
Saumya Gupta*, **Xiaoling Hu***, James Kaan, Michael Jin, Mutshipay Mpoy, Katherine Chung, Gagandeep Singh, Mary Saltz, Tahsin Kurc, Joel Saltz, Apostolos Tassiopoulos, Prateek Prasanna, Chao Chen
*European Conference on Computer Vision (**ECCV**)*, 2022 (**Oral Presentation, acceptance rate 2.7%**)

[23] **Trigger Hunting with a Topological Prior for Trojan Detection**
Xiaoling Hu, Xiao Lin, Michael Cogswell, Yi Yao, Susmit Jha, Chao Chen
*International Conference on Learning Representations (**ICLR**)*, 2022

[24] **A Manifold View of Adversarial Risk**
Wenjia Zhang, Yikai Zhang, **Xiaoling Hu**, Mayank Goswami, Chao Chen, Dimitris Metaxas
*International Conference on Artificial Intelligence and Statistics (**AISTATS**)*, 2022

[25] **Topology-Attention ConvLSTM Network for 3D Image Segmentation**
Jiaqi Yang*, **Xiaoling Hu***, Chao Chen, Chialing Tsai
*International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**)*, 2021

[26] **Topology-Aware Segmentation Using Discrete Morse Theory**
Xiaoling Hu, Yusu Wang, Li Fuxin, Dimitris Samaras, Chao Chen
*International Conference on Learning Representations (**ICLR**)*, 2021 (**Spotlight Presentation, acceptance rate 5.6%**)

[27] **3D Topology-Preserving Segmentation with Compound Multi-Slice Representation**
Jiaqi Yang*, **Xiaoling Hu***, Chao Chen, Chialing Tsai
*IEEE International Symposium on Biomedical Imaging (**ISBI**)*, 2021

[28] **Topology-Preserving Deep Image Segmentation**
Xiaoling Hu, Li Fuxin, Dimitris Samaras, Chao Chen
*Thirty-Third Conference on Neural Information Processing Systems (**NeurIPS**)*, 2019

Preprints

[1] **Learning to Upscale 3D Segmentations in Neuroimaging**
Xiaoling Hu, Peirong Liu, Dina Zemlyanker, Jonathan Williams Ramirez, Oula Puonti, Juan Eugenio Iglesias
Tech Report, 2025

[2] **Topo-R1: Detecting Topological Anomalies via Vision-Language Models**
Meilong Xu, Qingqiao Hu, **Xiaoling Hu**, Shahira Abousamra, Xin Yu, Weimin Lyu, Kehan Qi, Dimitris Samaras, Chao Chen
Tech Report, 2026

[3] **Unrolled Networks are Conditional Probability Flows in MRI Reconstruction**
Kehan Qi, Saumya Gupta, **Xiaoling Hu**, Qingqiao Hu, Weimin Lyu, Chao Chen
Tech Report, 2025

[4] **RB-FT: Rationale-Bootstrapped Fine-Tuning for Video Classification**
Meilong Xu, Di Fu, Jiaxing Zhang, Gong Yu, Jiayu Zheng, **Xiaoling Hu**, Dongdi Zhao, Feiyang Li, Chao Chen, Yong Cao
Tech Report, 2025

[5] **LoC-Path: Learning to Compress for Pathology Multimodal Large Language Models**
Qingqiao Hu, Weimin Lyu, Meilong Xu, Kehan Qi, **Xiaoling Hu**, Saumya Gupta, Jiawei Zhou, Chao Chen
Tech Report, 2025

[6] **Uncertainty Estimation for Pretrained Medical Image Registration Models via Transformation Equivariance**
Lin Tian, **Xiaoling Hu**, Juan Eugenio Iglesias
Tech Report, 2025

[7] ***Bézier Meets Diffusion*: Robust Generation Across Domains for Medical Image Segmentation**
Chen Li, Meilong Xu, **Xiaoling Hu**, Weimin Lyu, Chao Chen
Tech Report, 2025

[8] **Adversarial Vessel-Unveiling Semi-Supervised Segmentation for Retinopathy of Prematurity Diagnosis**
Gozde Merve Demirci, Jiachen Yao, Ming-Chih Ho, **Xiaoling Hu**, Wei-Chi Wu, Chao Chen, and Chia-Ling Tsai
Tech Report, 2024

[9] **Deep Statistic Shape Model for Myocardium Segmentation**
Xiaoling Hu, Xiao Chen, Terrence Chen, Shanhui Sun
Tech Report, 2022

Selected Honors and Awards

- CVPR Oral (Top 1%) 2025
- Catacosinos Fellowship (2 out of 200+ PhD students in SBU CS Department) 2023
- ICLR Spotlight (Notable-top-25% of accepted papers) 2023
- ECCV Oral (Top 2.7%) 2022
- ICLR Spotlight (Top 5.6%) 2021
- NeurIPS Travel Award 2019
- First-class Scholarship, Tsinghua University 2016

Mentoring

- Chen Li (**ICLR'23, ICCV'23, MICCAI'24**), Ph.D. student at Department of BMI, Stony Brook University
  Since Fall 2021
- Meilong Xu (**ECCV'24, CVPR'25 Oral Presentation, NeurIPS'25**), Ph.D. student at Department of CS, Stony Brook University
  Since Summer 2023
- Wentao Huang (**MICCAI'24**, **CVPR'26**, **TMI major revision**), Ph.D. student at Department of CS, Stony Brook University
  Since Summer 2023
- Qingqiao Hu, Ph.D. student at Department of CS, Stony Brook University
  Since Fall 2024
- Kehan Qi, Ph.D. student at Department of BMI, Stony Brook University
  Since Fall 2025
- Jiaqi Yang (**MICCAI'21, ISBI'21, MedIA'24, JBHI'25**), Ph.D. student at Department of CS, CUNY
  Spring 2020 − Summer 2023
- Saumya Gupta (**ECCV'22 Oral Presentation, NeurIPS'23**), Ph.D. student at Department of CS, Stony Brook University
  Fall 2021 − Summer 2023
- Mahmudul Hasan (**MICCAI'24**), Ph.D. student at Department of CS, Stony Brook University
  Summer 2023 − Summer 2024
- Luca Drole, Master student in BME, ETH Zurich
  Spring - Fall 2025

- John Xie, High School student → University of Michigan
Summer 2021

Professional Service

Organizer

- MICCAI'25 workshop on *Topology- and Graph-Informed Imaging Informatics (TGI3)* 2025
- MICCAI'24 workshop on *The First Workshop on Topology- and Graph-Informed Imaging Informatics (TGI3)* 2024
- MICCAI'23 tutorial on *Topology-Driven Image Analysis* 2023

Area Chair

- International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) 2025, 2026
- Conference on Neural Information Processing Systems (NeurIPS) 2025, 2026
- The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026
- International Conference on Artificial Intelligence and Statistics (AISTATS) 2026

Reviewing

- International Conference on Machine Learning (ICML) Since 2022
- International Conference on Learning Representations (ICLR) Since 2022
- Conference on Neural Information Processing Systems (NeurIPS) 2021 - 2024
- Computer Vision and Pattern Recognition (CVPR) 2021 - 2025
- European Conference on Computer Vision (ICCV) Since 2021
- European Conference on Computer Vision (ECCV) Since 2022
- Winter Conference on Applications of Computer Vision (WACV) Since 2022
- International Conference on Artificial Intelligence and Statistics (AISTATS) 2022 - 2025
- Learning on Graphs Conference (LoG) 2022 - 2025
- Medical Imaging with Deep Learning (MIDL) Since 2022
- AAAI Conference on Artificial Intelligence (AAAI) Since 2022
- International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) 2020 - 2024

- Transactions on Machine Learning Research (TMLR)
- IEEE Transactions on Medical Imaging (TMI)
- Medical Image Analysis (MedIA)

Talks

Principled Learning for Medical AI: Structure, Reliability, and Interpretability

- HIT Webinar - *Healthcare, Intelligence, Technology* — Aug. 2025
- Department of CSE, University of North Texas — Nov. 2025

***Learn2Synth*: A Learnable Data Synthesis Strategy for Image Segmentation**

- Nobrainer Seminar, Massachusetts Institute of Technology — June 2024

Deep Structural Reasoning for Biomedical Imaging

- School of CAI, Arizona State University — Feb. 2024

Topology-Aware Deep Image Segmentation

- MICCAI'23 tutorial on *Topology-Driven Image Analysis*, Vancouver — Oct. 2023

Learning Topological Representations for Deep Image Understanding

- Department of CS, Florida State University — Apr. 2023
- Department of BMI, Ohio State University — Mar. 2023
- Department of CS, Rochester Institute of Technology — Feb. 2023
- Department of ECE, University of California, Riverside — Feb. 2023
- Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital & Harvard Medical School — Nov. 2022

Learning Probabilistic Topological Representations Using Discrete Morse Theory

- Medical Imaging meets NeurIPS Workshop, New Orleans — Dec. 2022

Topology-Informed Image Analysis

- Center for Computational Neuroscience, Flatiron Institute — Oct. 2022

Topology-Aware Deep Image Segmentation

- Geometry and Topology meet Data Analysis and Machine Learning — Aug. 2021

Topology-aware Segmentation Using Discrete Morse Theory

- International Conference on Learning Representations (ICLR) — May 2021

Industry Experiences

- **Allen Institute, USA** — May 2022 - Aug. 2022
  Research Intern
  Mentor: *Dr.* Matheus Viana
  Topic: Topology-Aware Image Segmentation
- **United Imaging Intelligence (UII), USA** — May 2021 - Aug. 2021
  Research Intern
  Mentor: *Dr.* Shanhui Sun
  Topic: Deep Shape Model Based Network

- **Tencent Youtu Lab, China** Jun. 2017 - Jan. 2018
  Research Intern
  Mentor: *Dr.* Yuwing Tai
  Topic: Clothes Detection, Attribute Prediction

References

- **Chao Chen**
  Associate Professor, Stony Brook University
  chao.chen.1@stonybrook.edu
  `https://chaochen.github.io/`
- **Juan Eugenio Iglesias**
  Associate Professor, Massachusetts General Hospital & Harvard Medical School
  jiglesiasgonzalez@mgh.harvard.edu
  `https://lemon.martinos.org/pi/`
- **Bruce Fischl**
  Professor, Massachusetts General Hospital & Harvard Medical School
  bfischl@mgh.harvard.edu
  `https://scholar.google.com/citations?user=t7mytXkAAAAJ&hl=en`
- **Dimitris Samaras**
  SUNY Empire Innovation Professor, Stony Brook University
  samaras@cs.stonybrook.edu
  `https://www3.cs.stonybrook.edu/~samaras/`
- **Fuxin Li**
  Associate Professor, Oregon State University
  fuxin.li@oregonstate.edu
  `https://web.engr.oregonstate.edu/~lif/`
- **Prateek Prasanna**
  Assistant Professor, Stony Brook University
  prateek.prasanna@stonybrook.edu
  `https://you.stonybrook.edu/imaginelab/`